Diving Into DFIR

Read between the bytes

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As we all know, quarantine hasn't ended. And web-series have been a great companion. One of the trending ones, Paatal Lok, had a scene **spoiler alert** where one of the characters throws his phone in a lake, but eventually the police, as always, was able to dig crucial, plot-forwarding info from that phone.

But are these things even possible? And if yes, how in the world is data retrieval possible from a phone where the hard drive is clogged with water? But dear reader, that's just tip of the iceberg. Let's take a closer look at the iceberg that is DFIR.

WHAT DFIR?

Digital Forensics and Incident Response (DFIR, no idea how to pronounce the acronym, send us your versions!) is part of the computer forensics branch focused on the analysis of digital components, so as to uncover details of some events to determine whether some illegal event has occurred.

Now, DFIR can be bifurcated further into two subcategories: DF and IR. Digital Forensics (DF) is concerned with the evidence while digging through disk images, malware samples, network packet captures and what not. They are after bits and pieces and any other valuable information that can help them build the scenarios that lead to the compromise or the cybercrime event and uncover the bad actor. Incident response (IR) is the "brothers in arms" of Digital Forensics. When an incident is identified, a team of specialists aptly named "incident responders" are among the people who discover the problem behind the incident, try to mitigate the damages and prevent future attacks. Digital forensics is what comes afterwards.

DFIR plays a huge role in any business and is especially a crucial part of law enforcement. For example, think about a malware attack. Incident responders will first air gap the affected system/network and isolate it from the entire system to minimise the damage. Then comes the analysis part, having two main components: analysing the system state at the time of compromise and before, and analysing the malware itself. Numerous tools and kits are available specially catering to the needs of a DFIR specialist.

Now that you are introduced to the concept of DFIR, it's time to get into the technicalities.

DIGITAL FORENSICS

As they say, the devil is in the details. And the details matter a lot in digital forensics. DF consists of mainly two stages: acquisition and analysis. Let's take a look at them one by one.

Acquisition

The first step is to gather all the affected artefacts/exhibits, then you dig into the entrails. This can mean acquiring an SD card from a criminal's phone or a hard disk drive (someone could've tried to destroy, but forensic researchers can still retrieve data). Not the most technical part, but still an important stage. The interesting part is, an exhibit